Sec: OSR.IIT_*COSC(MODEL-A&APEX) | GTA-10(P2) | Date: 30-04-26
Time: 3 Hrs | 2021_P2 | Max. Marks: 180

KEY SHEET

MATHEMATICS

1	**ABCD**	2	**ABC**	3	**BCD**	4	**ABC**	5	**AC**
6	**B**	7	**343**	8	**48**	9	**11**	10	**0.07 TO 0.08**
11	**0.15**	12	**0.30**	13	**C**	14	**A**	15	**C**
16	**C**	17	**2**	18	**2**	19	**1330**		

PHYSICS

20	**AC**	21	**A**	22	**AD**	23	**BC**	24	**CD**
25	**AC**	26	**47.50 TO 47.70**	27	**34.55 TO 34.65**	28	**2.42 TO 2.44**	29	**8**
30	**1.44**	31	**9**	32	**A**	33	**C**	34	**A**
35	**C**	36	**9**	37	**80**	38	**4**		

CHEMISTRY

39	**BD**	40	**AD**	41	**D**	42	**ABD**	43	**ABCD**
44	**AD**	45	**10.50**	46	**8**	47	**3**	48	**3**
49	**9**	50	**4**	51	**B**	52	**C**	53	**B**
54	**B**	55	**3**	56	**10**	57	**84**		

For More Join: @IITJEE_Advanced

SOLUTIONS

MATHS

1. $\log_{x^2}16+\log_{2x}64=3$ Equation is defined when $x>0$ and $x\neq\frac{1}{2},1$

Now, $\frac{1}{2}\log_x 4^2+\log_{2x}4^3=3$

$\Rightarrow \log_x 4+3(\log_{2x}4)=3 \Rightarrow 2\log_x 2+6\log_{2x}2=3 \Rightarrow \frac{2}{(\log_2 x)}+\frac{6}{(\log_2 2x)}=3 \Rightarrow \frac{2}{t}+\frac{6}{(1+t)}=3\{$

where $t=\log_2 x\}$

$\Rightarrow 3t(t-2)+1(t-2)=0 \Rightarrow t=2,-\frac{1}{3} \Rightarrow \log_2 x=2,-\frac{1}{3} \Rightarrow x=4,2^{\frac{-1}{3}}$

2. $\left[f(x)=\cos 2x \text{ and } g(x)=e^x\right]$

3. $y=\sqrt[3]{6xy^2-11x^2y+6x^3}\ \ y^3=6xy^2-11x^2y+6x^3\ \ y^3-6xy^2+11x^2y-6x^3=0$

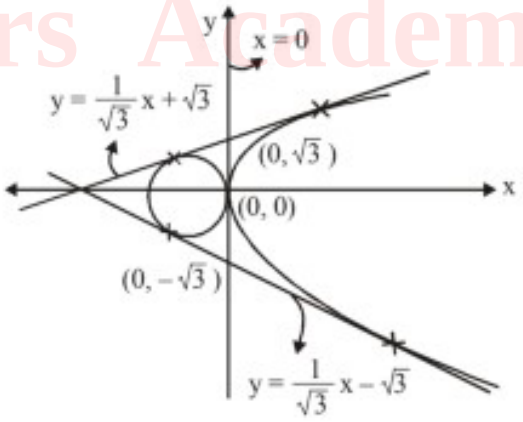
$\Rightarrow (y-x)(y-2x)(y-3x)=0 \Rightarrow y=x,y=2x,y=3x\ \therefore \frac{dy}{dx}=1,2,3\Big]$

4. Any tangent of slope ' m ' to $y^2=4x$ is

$y=mx+\frac{1}{m}$ or $mx-y+\frac{1}{m}=0$

If line (1) is also tangent to circle $(x+1)^2+y^2=1$

then using condition of tangency, we gets



$\left(-m+\frac{1}{m}\right)^2=(1+m^2) \Rightarrow (1-m^2)^2=m^2(1+m^2) \Rightarrow 1-2m^2=m^2 \Rightarrow 3m^2=1\ \therefore m=\pm\frac{1}{\sqrt{3}}$ Also, the line $x=0$ (called y -axis) is one of their common tangent.

5. The equation of tangent at P on ellipse is

$\frac{x}{3}\cos\theta+\frac{y}{2}\sin\theta=1$ So, $A\left(\frac{3}{\cos\theta},0\right),B\left(0,\frac{2}{\sin\theta}\right)$

Now, an ($\triangle OAB$)

$=\frac{1}{2}\left|\left(\frac{3}{\cos\theta}\right)\cdot\left(\frac{2}{\sin\theta}\right)\right|=\left|\frac{6}{\sin 2\theta}\right|\geq 6 \Rightarrow$ Minimum area of $\triangle OAB=6$ ans.

Also, $AB=\sqrt{9\sec^2\theta+4\operatorname{cosec}^2\theta}$

For More Join: @IITJEE_Advanced

$=\sqrt{9\left(1+\tan^2\theta\right)+4\left(1+\cot^2\theta\right)}=\sqrt{13+(3\tan\theta-2\cot\theta)^2+12}=\sqrt{25+(3\tan\theta-2\cot\theta)^2}\geq 5\Rightarrow$

Minimum length AB is 5

6. $2xy\frac{dy}{dx}=\left(x^2+y^2\right)\Rightarrow\left(2xydy-y^2dx\right)=x^2dx$

$\Rightarrow\left(\frac{2xydy-y^2dx}{x^2}\right)=dx\Rightarrow\int d\left(\frac{y^2}{x}\right)=\int dx\Rightarrow\frac{y^2}{x}=x+c$ As, $P(2,1)$ satisfy it, so

$\Rightarrow\frac{1}{2}=2+c\Rightarrow c=\frac{-3}{2}\Rightarrow y^2=x^2-\frac{3}{2}x\Rightarrow x^2-\frac{3}{2}x-y^2=0\Rightarrow\frac{\left(x-\frac{3}{4}\right)^2}{1}-\frac{y^2}{1}=\frac{9}{16}$, which is equation of rectangular hyperbola, whose eccentricity $=\sqrt{2}$

7. $AB=I=BA\ A(B+A)=2I\ AB+A^2=2I\Rightarrow A^2=I\Rightarrow A=A^{-1}=B\ |A|=\pm 1$ (i) det.

$\left(4(\text{ adj.}A)^2+3(\text{ adj.}B)^2\right)=\text{ det. }\left(7(\text{ adj.}A)^2\right)$

$=7^3.|(\text{ adj.}A)^2|=343|A|^4=343$

8. $P=\left(A^{-1}+B^{-1}\right)(\text{ adj. }(2A)+\text{ adj. }(2B))$

$=(2A)(2\text{ adj. }(2A))=2(2A)\text{ adj. }(2A)$

$=2.2^3|A|I=\pm 16I\ \therefore \text{Tr}(P)=\pm 48$

9. $f(x)=x^3-6\sin\theta\cdot x^2+\left(2\sin^2\theta+q\right)x-1\ f'(x)=3x^2-12\sin\theta.x+2\sin^2\theta+q$ For exactly one tangent to be drawn to the graph of $y=f(x)$, parallel to $y=x$.

$f'(x)=1$ should have one root

$D=0\Rightarrow q=1+10\sin^2\theta$ .. largest value of q is 11. Ans.

10. Smallest value of $q=1$ when $\sin\theta=0$

Now, $f(x)=x^3+x-1$

$f(2)=9\Rightarrow f^{-1}(9)=2\left.\frac{d}{dx}\left(f^{-1}(x)\right)\right|_{x=9}=\frac{1}{f'(2)}=\frac{1}{13}\left\{\because f(x)=3x^2+1\right\}\Big]$

11. $P(A\cup B)=P(A)+P(B)-P(A\cap B)\ \frac{3}{5}=\frac{1}{4}+\frac{1}{2}-P(A\cap B)\Rightarrow P(A\cap B)=\frac{3}{20}=0.15$

12. $P(A/B)=\frac{P(A\cap B)}{P(B)}=\frac{3/20}{1/2}=\frac{3}{10}=0.30$

$\frac{[x]^2}{4}=x\cdot\{x\}\ [x]^2=4(\{x\}+[x])\{x\}\Rightarrow[x]^2=4\{x\}[x]+4\{x\}^2\ 4\{x\}^2+4\{x\}[x]-[x]^2=0$

$\left(\frac{2\{x\}}{[x]}\right)^2+2\left(\frac{2\{x\}}{[x]}\right)-1=0;\ \text{let }\frac{2\{x\}}{[x]}=r$

$r^2+2r-1=0\Rightarrow r=\frac{-2\pm\sqrt{4+4}}{2}=\frac{-2\pm 2\sqrt{2}}{2}=-1\pm\sqrt{2}\ \therefore r=\sqrt{2}-1\{\because x>0\Rightarrow r>0\}$

$\frac{2\{x\}}{[x]}=\sqrt{2}-1\Rightarrow\{x\}=\left(\frac{\sqrt{2}-1}{2}\right)[x]\ 0<\left(\frac{\sqrt{2}-1}{2}\right)[x]<1\Rightarrow 0<[x]<\frac{2}{\sqrt{2}-1}\Rightarrow 0<[x]<\underbrace{2\left(\sqrt{2}+1\right)}_{4.14}\therefore$

Possible values of $[x]$ are 1,2,3 and 4

13. $\frac{[x]^2}{4}=x\cdot\{x\}$

For More Join: @IITJEE_Advanced

$\left[x\right]^2 = 4\left(\{x\}+\left[x\right]\right)\{x\} \Rightarrow \left[x\right]^2 = 4\{x\}\left[x\right]+4\{x\}^2\ 4\{x\}^2+4\{x\}\left[x\right]-\left[x\right]^2 = 0$

$\left(\frac{2\{x\}}{\left[x\right]}\right)^2 + 2\left(\frac{2\{x\}}{\left[x\right]}\right)-1=0;\ \text{let}\ \frac{2\{x\}}{\left[x\right]} = r$

$r^2+2r-1=0 \Rightarrow r = \frac{-2\pm\sqrt{4+4}}{2} = \frac{-2\pm 2\sqrt{2}}{2} = -1\pm\sqrt{2}\ \therefore r = \sqrt{2}-1\{\because x>0 \Rightarrow r>0\}$

$\frac{2\{x\}}{\left[x\right]} = \sqrt{2}-1 \Rightarrow \{x\} = \left(\frac{\sqrt{2}-1}{2}\right)\left[x\right]\ 0<\left(\frac{\sqrt{2}-1}{2}\right)\left[x\right]<1 \Rightarrow 0<\left[x\right]<\frac{2}{\sqrt{2}-1} \Rightarrow 0<\left[x\right]<\underbrace{2\left(\sqrt{2}+1\right)}_{4.14}\therefore$

Possible values of $\left[x\right]$ are 1,2,3 and 4

14. $1<x<2 \Rightarrow \left[x\right]=1\ \{x\} = \frac{\sqrt{2}-1}{2} \Rightarrow x = \frac{\sqrt{2}-1}{2}+1 = \frac{\sqrt{2}+1}{2}\therefore$ Required product is,

$\frac{\sqrt{2}+1}{2}\cdot\frac{1}{2}\cdot\frac{\sqrt{2}-1}{2} = \frac{1}{8}$

15. $\vec{v}_1 = (\sin\theta)\hat{i}+(\cos\theta)\hat{j}+(a-3)\hat{k}$

$\vec{v}_2 = (\sin\theta+\cos\theta)\hat{i}+(\cos\theta-\sin\theta)\hat{j}+(b-4)\hat{k}\ \vec{v}_3 = (\cos\theta)\hat{i}-(\sin\theta)\hat{j}+(c-5)\hat{k}$ (i) Given,

$\vec{v}_1+\vec{v}_2+\vec{v}_3 = \lambda\hat{i}$

$\Rightarrow (2\sin\theta+2\cos\theta)\hat{i}+(2\cos\theta-2\sin\theta)\hat{j}+(a+b+c-12)\hat{k} = \lambda\hat{i}\ \therefore 2\cos\theta-2\sin\theta=0$ and $a+b+c-12=0$

$\Rightarrow \tan\theta = 1$ and $a+b+c=12; a,b,c\in N$

$\theta\in[0,2\pi]\ \therefore a+b+c=9 \Rightarrow {}^{11}C_2 = 55\ \therefore \theta = \frac{\pi}{4}$ or $\frac{5\pi}{4}$

$\therefore$ Number of quadruplets (a,bc,θ) are $55\times 2 = 110$

16. Given $\vec{v}_1,\vec{v}_2,\vec{v}_3$ represents of $\triangle PQR$ in xy-plane

$\therefore$ coefficient of k in all 3 vector $=0$

Hence, $a=3,\ b=4$ and $c=5$

Also, triangle PQR is right triangle

$\Delta PQR = \frac{1}{2}\times 1\times 1 = \frac{1}{2}$ and $\triangle ABC = \frac{1}{2}\times 3\times 4 = \frac{12}{2}$

$\therefore \frac{\triangle ABC}{\triangle PQR} = \frac{\frac{12}{2}}{\frac{1}{2}} = 12$

17. $\sum_{n=0}^{\infty}\tan^{-1}\left(\frac{\tan^{-1}\left(\frac{(n+2)-(n+1)}{1+(n+1)(n+2)}\right)}{1+\tan^{-1}\left(\frac{1}{n+1}\right)\cdot\tan^{-1}\left(\frac{1}{n+2}\right)}\right)$

For More Join: @IITJEE_Advanced

$$== \sum_{n=0}^{\infty} \tan^{-1}\left(\frac{\tan^{-1}\left(\frac{1}{(n+1)}\right)-\tan^{-1}\left(\frac{1}{n+2}\right)}{1+\tan^{-1}\left(\frac{1}{n+1}\right)\cdot\tan^{-1}\left(\frac{1}{n+2}\right)}\right) = \sum_{n=0}^{\infty}\left(\tan^{-1}\left(\tan^{-1}\left(\frac{1}{n+1}\right)\right)-\tan^{-1}\left(\tan^{-1}\left(\frac{1}{n+2}\right)\right)\right)$$

$$= \tan^{-1}\left(\tan^{-1}(1)\right) = \tan^{-1}\left(\frac{\pi}{4}\right)$$

$$\therefore p = \pi$$

$$\left[\cos^{-1}\left(\cos(\pi-1)\right)\right] = [\pi-1] = 2$$

18. $f(x) = \sqrt{6-5^{\sin^{-1}x}-5^{1-\sin^{-1}x}}$ and $g(x) = \sqrt{5^{\sin^{-1}x}}$

$x \in [-1,1]$

$$6-5^{\sin^{-1}x}-5^{1-\sin^{-1}x} = 6-5^{\sin^{-1}x}-\frac{5}{5^{\sin^{-1}x}}$$

$$= -\left[\frac{\left(5^{\sin^{-1}x}\right)^2-6\left(5^{\sin^{-1}x}\right)+5}{5^{\sin^{-1}x}}\right]$$

$$= \frac{4-\left(5^{\sin^{-1}x}-3\right)^2}{5^{\sin^{-1}x}}$$

$$\Rightarrow y = h(x) = f(x)\cdot g(x) = \sqrt{4-\left(5^{\sin^{-1}x}-3\right)^2}$$

for domain $1 \le 5^{\sin^{-1}x} \le 5 \Rightarrow \sin^{-1}x \in [0,1]$

$$\Rightarrow 0 \le 4-\left(5^{\sin^{-1}x}-3\right)^2 \le 4 \Rightarrow y \in [0,2]$$

19. $$\sum_{r=0}^{19} r(r+1)\frac{{}^{20}C_{r+1}}{{}^{20}C_r} = \sum_{r=0}^{19} r(r+1)\frac{(20-(r+1)+1)}{(r+1)}$$

$$= \sum_{r=0}^{19} r\cdot(20-r) = 1330$$

PHYSICS

20. Resolving power, $RP = \frac{2\mu\sin\alpha}{\lambda}$

minimum separation between objects required

$$\Delta x = \frac{\lambda}{2\mu\sin\alpha}$$

For $\lambda_0 = 480$ nm :

$$RP = \frac{2\times1.6\sin74^\circ}{480\times10^{-9}} = \frac{2\times1.6\times\frac{24}{25}}{480\times10^{-9}} = 6.4\times10^6$$

$$\Delta x = \frac{480\times10^{-9}}{2\times1.6\times\frac{24}{25}}\ m = 1.5625\times10^{-7}$$

$$m = 0.15625\mu\ m$$

For $\lambda_0 = 420$ nm

$$RP = 7.31\times10^6$$

For More Join: @IITJEE_Advanced

$\Delta x = 0.14\mu$ m

$\Rightarrow$ Filaments separated by 0.18μ m can be resolved by both 420 mm & 480 mm when switching from 480 nm to 420 nm , the RP increases by a factor of $\frac{8}{7} \Rightarrow \%$ improvement $= 14.29\%$

For 480 nm and air

$\Delta x = 1.6 \times 0.15625\mu$ m $= 0.25\mu$ m

21. $\beta = \frac{\lambda D}{d} \Rightarrow \lambda = \frac{\beta d}{D}$

$\frac{\Delta\lambda}{\lambda} = \frac{\Delta\beta}{\beta} + \frac{\Delta d}{d} + \frac{\Delta D}{D}$

$D = 1500$ mm, $\Delta D = 1$ mm

$d = 0.33$ mm, $\Delta d = 0.01$ mm

$\beta = 2.50$ mm, $\Delta\beta = 0.01$ mm

$\Rightarrow \lambda = \frac{2.5 \times 0.33}{1500}$ mm $= 550$ nm

$\frac{\Delta\lambda}{\lambda} = \frac{0.01}{2.5} + \frac{0.01}{0.33} + \frac{1}{1500}$

$\Delta\lambda = 19.23$ nm

22. $m\omega_0 r_0^2 = m\omega r^2$

$\Rightarrow \omega = \omega_0 \left(\frac{r_0}{r}\right)^2$

$mg(r_0 - r) + \frac{1}{2}m(\omega_0 r_0)^2 = \frac{1}{2}m(\omega r)^2 + \frac{1}{2} \times 2m \times \dot{r}^2$

at maximum/minimum $r, \dot{r} = 0$

$\Rightarrow mg(r_0 - r) + \frac{1}{2} m\omega_0^2 r_0^2 = \frac{1}{2} m\omega^2 r^2 + 0$

$2g(r_0 - r) = \omega_0^2 \frac{r_0^4}{r^2} - \omega_0^2 r_0^2$

$2g(r_0 - r).r^2 = \omega_0^2 r_0^2 (r_0^2 - r^2)$

$2gr^2 - \omega_0^2 r_0^2 (r_0 + r)$

$2gr^2 - \omega_0^2 r_0^2 r - \omega_0^2 r_0^3 = 0$

$r = \frac{\omega_0^2 r_0^2 + \sqrt{\omega_0^4 r_0^4 + 8g\omega_0^2 r_0^3}}{4g}$

$= \frac{\omega_0^2 r_0^2}{4g}\left(1 + \sqrt{1 + \frac{8g}{\omega_0^r r}}\right)$

$= \frac{r_0}{12}\left(1 + \sqrt{1+24}\right) = \frac{r_0}{2}$

$mg - T = m\left(\frac{mg - m\omega^2 r}{2m}\right)$

$T = mg - \frac{1}{2} m(g - \omega^2 r)$

For More Join: @IITJEE_Advanced

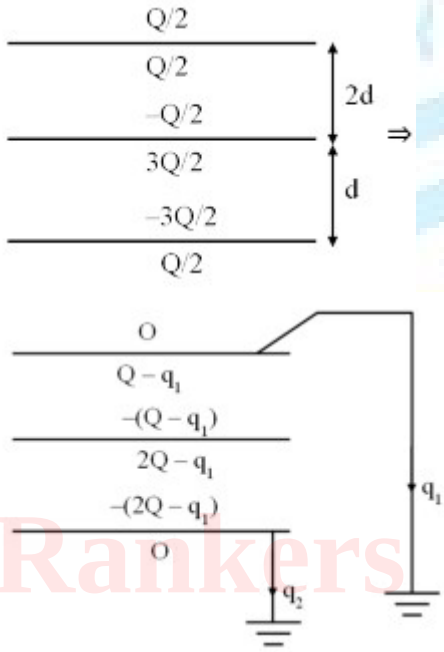
$$= \frac{mg}{2}\left(1+\frac{\omega_0^2}{g}\frac{r_0^4}{r^4}r\right)$$

$$= \frac{mg}{2}\left(1+\frac{1}{3}\cdot\left(\frac{r_0}{r}\right)^3\right)$$

T is maximum when r is minimum

$$\Rightarrow T_{max} = \frac{mg}{2}\left(1+\frac{1}{3}\times 8\right) = \frac{11}{6}mg$$

23. Correct Answer is (B,C)
24. Correct Answer is (C,D)
25.



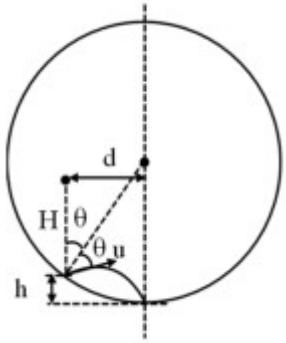
$$q_1 + q_2 = Q$$

and $$\frac{Q-q_1}{\epsilon_0}\cdot 2\,d + \frac{2Q-q_1}{\epsilon_0}\cdot d = 0$$

$$2Q - 2q_1 + 2Q - q_1 = 0$$

$$\Rightarrow q_1 = \frac{4}{3}Q; q_2 = -\frac{Q}{3}$$

26. Correct Answer is 47.50 to 47.70
27. Correct Answer is 34.55 to 34.65
28. Correct Answer is 2.42 to 2.44
29.



For More Join: @IITJEE_Advanced

$h = R - \sqrt{R^2 - d^2}$

$\sin\theta = \frac{d}{R}, \cos\theta = \frac{R-h}{R},$

$\sin 2\theta = \frac{2\,d(R-h)}{R^2}, \cos 2\theta = 2\left(\frac{R-h}{R}\right)^2 - 1$

$u = \sqrt{2gH}\ \cot 2\theta = \frac{2(R-h)^2 - R^2}{2\,d(R-h)}$

$-h = u\cos 2\theta . T - \frac{1}{2}gT^2; d = u\sin 2\theta . T$

$T = \frac{d}{u\sin 2\theta} = \frac{R^2}{24(R-h)}$

$= \frac{R^2}{2\sqrt{2gH}(R-h)} = \frac{R^2}{\sqrt{gH}(R-h)}\frac{1}{2\sqrt{2}} \Rightarrow \alpha$

$= 2\sqrt{2}$

also, $-h = d\cdot\cot 2\theta - \frac{1}{2}\frac{gd^2}{u^2\sin^2 2\theta}$

$-h = d\cdot\frac{2(R-h)^2 - R^2}{2(R-h)} - \frac{1}{2}\frac{gd^2R^4}{u^2\cdot 4\,d^2(R-h)^2}$

$-h = \frac{2(R-h)^2 - R^2}{2(R-h)} - \frac{gR^4}{8u^2(R-h)^2}$

$-h = (R-h) - \frac{R^2}{2(R-h)} - \frac{gR^4}{8u^2(R-h)^2}$

$R = \frac{R^2}{2(R-h)} + \frac{gR^4}{8u^2(R-h)^2};$

$1 - \frac{R}{2(R-h)} = \frac{gR^3}{8u^2(R-h)^2}$

$\frac{(R-2h)}{2(R-h)} = \frac{gR^3}{8u^2(R-h)^2};$

$u^2 = \left(\frac{gR^3}{4(R-h)(R-2h)}\right.$

$(R-h)(R-2h) = \frac{R^3}{8H}$

$2h^2 - 3Rh + R^2\left(1 - \frac{R}{8H}\right) = 0$

$h = \frac{3R \pm \sqrt{9R^2 - 8R^2\left(1 - \frac{R}{3H}\right)}}{4}$

$= \frac{R}{4}\left(3 - \sqrt{1 + \frac{R}{4}}\right)$

for $d \to 0,\ h \to 0$

For More Join: @IITJEE_Advanced

$$h=\frac{3R\pm\sqrt{9R^2-8R^2\left(1-\frac{R}{3u}\right)}}{4}$$

$$=\frac{R}{4}\left(3-\sqrt{1+\frac{R}{H}}\right)$$

for $d\to 0,\ h\to 0$

$$\Rightarrow 3=\sqrt{1+\frac{R}{H}}\Rightarrow\frac{R}{H}=8\Rightarrow\beta=8$$

30. $d=1\text{ mm}, D=1.2\text{ m}$

$\lambda_1=600\text{ nm}, \lambda_2=400\text{ nm}$

$$\beta_1=\frac{\lambda_1 D}{d}=0.72\text{ mm}$$

$$\beta_1=\frac{\lambda_2 D}{d}=0.48\text{ mm}$$

$n_1\beta_1=n_2\beta_2=y$

$$\frac{n_1}{n_2}=\frac{4}{6}=\frac{2}{3}\Rightarrow 3n_1=2n_2$$

$n_{1_{min}}=2$

$\Rightarrow y_{min}=2\beta_1=1.44\text{ mm}$

31. Correct Answer is 9

32. $$v=\int_a^c\frac{\lambda}{2\pi\epsilon_0 r}\cdot\frac{1}{k}\cdot dr+\int_c^b\frac{\lambda}{2\pi\epsilon_0 r}dr$$

$$=\frac{\lambda}{2\pi\epsilon_0}\left(\frac{1}{k}\ln\left(\frac{c}{a}\right)+\ln\left(\frac{b}{c}\right)\right)$$

$$c=\frac{\lambda.\ell}{v}=\frac{2\pi\epsilon_0\ell}{\frac{1}{k}\ln\left(\frac{c}{a}\right)+\ell\ln\left(\frac{b}{c}\right)}$$

$$=\frac{2\pi\epsilon_0\ell}{\frac{1}{k}\left(\ln\sqrt{\frac{b}{a}}+\ln\left(\sqrt{\frac{b}{a}}\right)\right)}$$

33. Let x length of dielectric is inside the capacitor, then

$$C(x)=\frac{4\pi\epsilon_0 x.k}{(1+k)\ln\left(\frac{b}{a}\right)}+\frac{2\pi\epsilon_0(\ell-x)}{\ln\left(\frac{b}{a}\right)}$$

$$=\frac{2\pi\epsilon_0}{\ell\ln\left(\frac{b}{a}\right)}\left(\ell-x+\frac{2kx}{1+k}\right)$$

$$\frac{dc}{dx}=\frac{2\pi\epsilon_0}{\ln\left(\frac{b}{a}\right)}\left(\frac{2k}{1+k}-1\right)=\frac{2\pi\epsilon_0(k-1)}{(k+1)\ln\left(\frac{b}{a}\right)}$$

If a charge Q is stored, EPE

$$U=\frac{Q^2}{2C}$$

For More Join: @IITJEE_Advanced

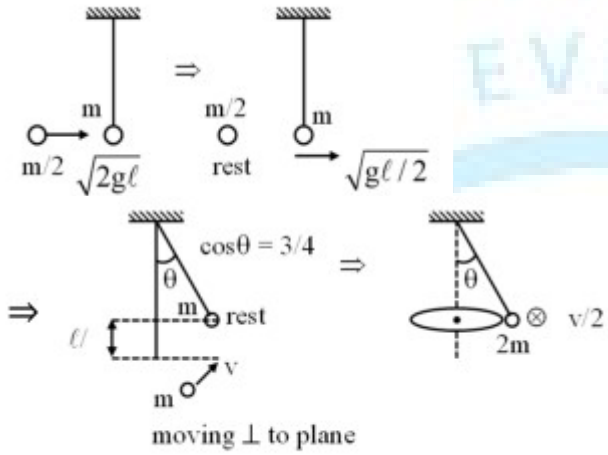
$$F = -\frac{dU}{dx} = \frac{Q^2}{2C^2}\frac{dC}{dx} = \frac{v^2}{2} \times \frac{2\pi\epsilon_0 (k-1)}{(k+1)\ln\left(\frac{b}{a}\right)}$$

$\Rightarrow$ work done in removing dielectric

$$W = F.\ell = \frac{\pi\epsilon_0 \ell (k-1) v^2}{(k+1)\ell n\left(\frac{b}{a}\right)}$$

34. Correct Answer is (A)

35.

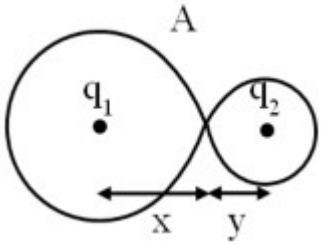


$$2\,m\frac{(v/2)^2}{\ell\sin\theta}\cdot\cos\theta = 2mg\cdot\sin\theta$$

$$\Rightarrow v^2 = \frac{4g\sin^2\theta\cdot\ell}{\cos\theta} = 4\,g\ell \times \frac{7/16}{3/4} = \frac{7\,g\ell}{3}$$

Time of flight, $T = \sqrt{\frac{2}{g}\times\left(\ell + \frac{\ell}{4}\right)} = \sqrt{\frac{5\ell}{2\,g}}$

36.



$$\frac{ka_1}{x} + \frac{ka_2}{\ell - x} = 900$$

and $\frac{kq_1}{x^2} - \frac{kq_2}{(\ell - x)^2} = 0$

$$\Rightarrow \frac{x^2}{4} = \frac{(\ell - x)^2}{1} \Rightarrow x = 2(\ell - x) \Rightarrow x = \frac{2\ell}{3}$$

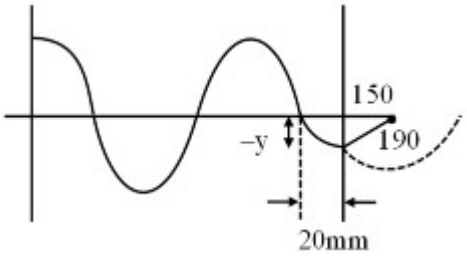
$$9\times10^9\left(\frac{4\times10^{-9}}{\frac{2\ell}{3}} + \frac{10^{-9}}{\frac{\ell}{3}}\right) = 900$$

For More Join: @IITJEE_Advanced

$$3 = \frac{100\ell}{3}$$

$$\Rightarrow \ell = \frac{9}{100}\ \text{m} = 9\ \text{cm}$$

37. Radine of circular path $= y_0 = 170$ mm



$$870\ \text{mm} = 5\times 170\ \text{mm} + 20\ \text{mm}$$

$$-y = \sqrt{170^2 - 150^2} = \sqrt{20\times 320} = 80\ \text{mm}$$

38. $A_y = 2.5$ cm

$$w_y A_y = v_0$$

$$\Rightarrow w_y = \frac{10\ \text{cm/s}}{2.5\ \text{cm}} = 4\ \text{rad/s}$$

CHEMISTRY

39. $XeF_2 + PF_5 \rightarrow [XeF]^+[PF_6]^-$

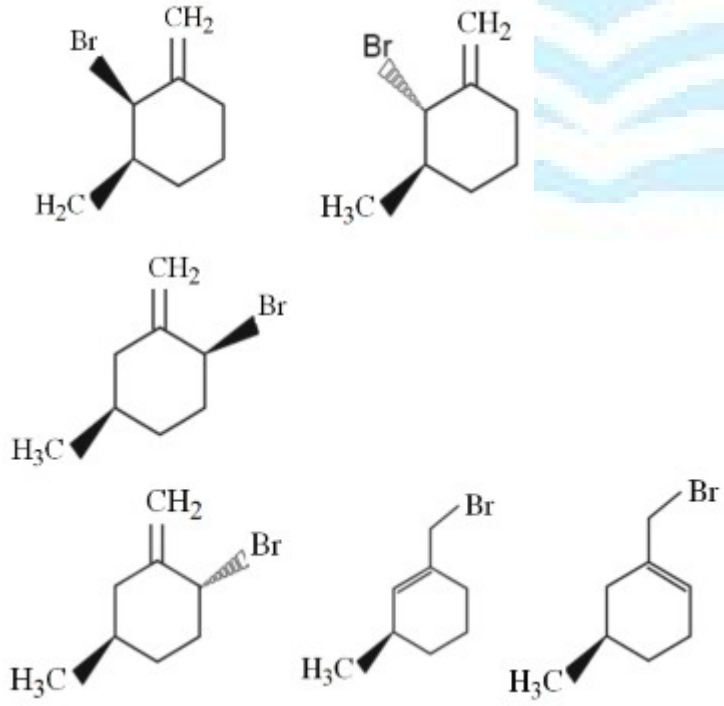
$XeF_4 + O_2\ F_2 \rightarrow XeF_6 + O_2$

$XeF_2 + H_2O \rightarrow Xe + HF + O_2$

$XeF_4 + 2H_2O \rightarrow \frac{2}{3}Xe + \frac{1}{3}XeO_3 + HF + \frac{1}{2}O_2$

$XeF_6 + H_2O \rightarrow XeOF_4 \rightarrow XeO_2\ F_2 \rightarrow XeO_3$

40.



For More Join: @IITJEE_Advanced

41. P is 5-ketohexanal
Q is 3-hydroxycyclohexanone
R is cyclohexanol
S is cyclohexanone
T is 2-cyclohexenone
U is cyclohexylbromide
V is cyclohexylamine

42. $MnO_4^- \xrightarrow{\text{conc. } H_2SO_4} Mn_2O_7$ (green)

$KI + Cl_2 \rightarrow I_2$ (violet)

$MnO_2 + 4HCl \rightarrow MnCl_2 + 2H_2O + Cl_2$

$N_2\ S_2O_3 + 4Cl_2 + 5H_2O \rightarrow 2NaHSO_4 + 8HCl$

43. Correct option answer is Abcd

44. Correct option answer is ad

45. $pH = \frac{PK_{a_2} + PK_{a_3}}{2} = 10.5$

46. $HPO_4^{2-} + H_2PO_4^- \ V = 40ml$

105 -

$5 - 5$

$pH = 8 + \log\frac{5}{5} = 8$

47. The correct answer is (3)

48. The correct answer is (3)

49. The correct answer is (9)

50. The correct answer is (4)

51.

$A_2B_3 \rightleftharpoons 2A_{(aq.)}^{3+} + 3B_{(aq.)}^{-2}$

$t = 0\ a_0$

$t = 0\ a_0 - x\ 2x\ 3x$

$\pi \propto C$, where C effective molarity

$\pi_0 \propto a_0$

$\pi \propto (a_0 + 4x)$

$x \propto [(\pi - a_0)/4]$

For More Join: @IITJEE_Advanced

$$K = \frac{1}{t}\ell n\left(\frac{a_0}{a_0 - x}\right)$$

$$= \frac{1}{t}\ell n\frac{a_0}{a_0 - \left(\frac{x-a_0}{4}\right)} \Rightarrow \frac{1}{t}\ell n\left[\frac{4a_0}{5a_0 - \pi}\right]$$

$$K = \frac{1}{10}\ell n\left(\frac{4\times 2}{5\times 2 - 6}\right) \Rightarrow \frac{1}{10}\ell n2$$

$$t_2 = \frac{1}{K}\ell n\left(\frac{4\times 2}{5\times 2 - x}\right)$$

$$20 = \frac{10}{\ell n2}\left(\frac{8}{10 - x}\right)$$

$x' = 8$ mm Hg

52. $K\times 10 = \ln 2$

$$K = \left(\frac{\ell n2}{10}\right) = \frac{\ell n2}{t_{1/2}}$$

$\Rightarrow t_{1/2} = 10$ min

$t_{0.875} = 3\times 10 = 30$ min

53. The correct answer is option (B)

54. The correct answer is option (B)

55. $Cd^{+2}, As^{+3}, As^{+5}$ give yellow ppt.

56. A sample can emit 10 different spectral lines when $n_2 = 5$ and $n_1 = 1$

maximum energy emitted is 117.5 eV

$$\Delta E = 13.6z^2\left(\frac{1}{1} - \frac{1}{25}\right) = 117.5\text{eV} \Rightarrow z = 3$$

Photon of minimum energy is emitted from $5 \rightarrow 4$ transition.

$$\Delta E = 13.6\times 9\times\left(\frac{1}{16} - \frac{1}{25}\right)$$

$$= \frac{13.6\times 9\times 9}{16\times 25} = 2.754\text{eV}$$

Work function of metal $= 1.254$eV

K.E.$_{max}$. of photo electrons $= 1.5$eV

de-Broglie wavelength of photo electron is

$$\lambda = \sqrt{\frac{150}{1.5}} = \sqrt{100}\ \text{A}^\circ$$

$\lambda = 10\ \text{A}^\circ$

57. $H_2O \rightarrow H_2 + \frac{1}{2}O_2\ (nf = 4)$

Moles of O_2 produced $= \frac{40\ F\times 0.75}{F\times 4} = 7.5$moles

$C_2H_4(g) + 3O_2(g) \rightarrow 2CO_2(g) + 2H_2O(\ell)$

moles $= \frac{3.5}{28} = \frac{1}{8}7.5$

The heat released by $\frac{1}{8}$ moles of C_2H_4 is

For More Join: @IITJEE_Advanced

$= C.\Delta T$

$= 3 \times 0.51.5KJ$

$\Delta U\left(KJ/mol\right) = \dfrac{-1.5 \times 28}{3.5} = -12KJ/mol$

$C_2H_4(g) + 3O_2(g) \rightarrow 2CO_2(g) + 2H_2O(\ell)$

$\Delta H = \Delta U + \Delta ngRT$ (for one mole)

$= -12 + \dfrac{(-2) \times 8 \times 300}{1000}$

$= -12 - 4.8 = -16.8KJ/mol.$

for 5 moles of ethene

$|\Delta H| = +16.8 \times 5 = 84KJ$

Rankers Academy JEE
THE NARAYANA GROUP

For More Join: @IITJEE_Advanced